

Jaehoon Song

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Software Engineer with experience building distributed systems, SOAP/REST APIs, databases, multi-threading/concurrency, and full-stack applications. Strong foundation in algorithms, systems programming, and software design principles/patterns.

EDUCATION

Georgia Institute of Technology | Atlanta, GA

Jan 2022 – Dec 2025

Bachelor of Science in Computer Science

GPA: 3.87/4.00

- Built a proof-based theoretical foundation through coursework in analysis, statistics, discrete mathematics, number theory, data structure & algorithms (DSA), automata (FSM) & complexity theory, and online algorithms
- Developed frontend development (UI/UX) experience: Android App development, D3.js data visualization, OpenGL, and Unity game development by Agile's Scrum framework for project management, object-oriented design patterns and principles, design documentation, and strong cross-team communication skills.
- Achieved Dean's List recognition, College of Computing, 2022

Gwinnett Technical College | Lawrenceville, GA

Jan 2018 – May 2021

Associate of Applied Science in Computer Science

GPA: 3.96/4.00

- Built a practical software engineering foundation integrating desktop, web, and mobile development through coursework in IT Analysis, Project Management, and Advanced Systems Project
- Developed expertise in operating system fundamentals and networking with C/C++, concurrency/multi-threading, memory management, device I/O, Web services (Servlets/JSP)
- Gained proficiency in application development: Java for enterprise systems and database management (Oracle SQL, MS Access, Apache Derby), Python for data analysis and media processing
- Earned Java Programmer Certificate, Honor's List, and Award of Merit

EXPERIENCE

Columbia Academy | Duluth, GA

Sep 2024 – Present

Freelance Lecturer (1099 Contractor)



- Instructed mathematics courses (AP and Georgia Tech Math), science courses (Physics), and computer science courses (AP CSA and principles in Java/Python/JavaScript) following the curriculum standards for high school students
- Designed coding challenge problems and implemented comprehensive unit tests to evaluate algorithmic correctness and programming proficiency
- Developed comprehensive SAT Summer bootcamp materials using customized L^AT_EX environments and commands for professional typesetting and released them as open educational resources

Georgia Institute of Technology, School of Physics | Atlanta, GA

Jan 2023 – Dec 2024

Teaching Assistant (TA)



- Operated lab sessions for PHYS 2211 under Emily Alicea-Muñoz
- Assisted students in completing lab assignments and understanding statics/dynamics in physics including thermodynamics and quantum physics
- Earned Faculty Honors from School of Physics, 2023

Georgia Institute of Technology, School of Computer Science | Atlanta, GA

May 2022 – Dec 2022

Undergraduate Teaching Assistant (UTA)



- Conducted office hours and recitation sessions for CS 2050 to support student learning in discrete mathematics (combinatorics, graph theory, number theory, and cryptography fundamentals) under the supervision of CoC faculty
- Prepared and edited problem sets, solution guides, and recitation handouts using L^AT_EX and coordinated with instructors

SK C&C USA | Suwanee, GA

Apr 2021 – Sep 2021

Software Engineer/Developer

- Contributed to the MES system for SK Global Battery manufacturing, developed in WinForms (.NET Framework), focusing on TCP/IP networking components (SOAP messaging over HTTP) and PLC communication using XML
- Maintained and managed VMs, Network configurations, and Oracle Database operations for data consistency across factory plant systems
- Designed user documentation to help factory plant workers understand and operate MES user interfaces effectively

SELECTED PROJECTS

JS Law Group Website | Remote
Freelance Web Developer (Contractor)

Oct 2025 – Present



- Developed a responsive bilingual (English/Korean) law firm website using Bootstrap and SCSS customization, optimized for cross-device compatibility
- Built modular JavaScript and specialized contact forms (general inquiry, personal injury, criminal defense) with validation, a lightweight i18n system with JSON locale files for multilingual support, and a QA debug mode

Greater Youth Collaborative Opus (GYCO) | Remote
Freelance Web Developer (Contractor)

May 2025 – Oct 2025



- Developed a complete static website for Korean-American youth orchestra institution following MVC architecture with SASS for Bootstrap customization and Vanilla JavaScript
- Designed build automation scripts, intuitive UI/UX with SNS integration, article management system, and content management capabilities for student administrators to understand web development principles and maintain the website independently

TradingBot (Automated Trading System) | Remote
Software Engineer/Developer (Solo Project)

Mar 2025 – May 2025



- Designed and implemented an automated trading system using low-level HTTP library (`request`) for direct API connectivity with brokers/exchanges using JSON data exchange format
- Built a scheduler-driven concurrency engine to continuously initiate and coordinate trading logic
- Implemented accounting python modules for real-time asset tracking, debit/credit (cash-to-equity) monitoring, rebalancing of index stock (SPY, QQQ, etc.), and dividend option strategies
- Built order management, error handling, logging, and reconciliation routines to maintain accounting integrity and auditability

Grady Health System (Not Publicly Available) | Atlanta, GA
Software Engineer/Developer

Sep 2023 – May 2024



- Developed a heart-care application to reduce hospital readmissions by detecting complications early from patient-reported data and following the Scrum framework of Agile as a year-long capstone project coordinated with Dr. Muling Lin from College of Medicine, Emory University
- Led system architecture design and comprehensive design documentation
- Developed a local database (Java, prototype only, build automation by Apache Ant) with minimal UI, integration with a RESTful Flask web server over HTTP, and socket connection for private network configuration/security using Apache Derby database engine to customize SQL rules for specific medical use cases
- Implemented statistical analysis modules (Python) for patient-reported data trends and alert systems for clinicians
- Delivered mobile app for patients and secure web portal for clinicians, with intellectual property assigned to College of Medicine, Emory University

LANGUAGES

English (Fluent), Korean (Native), Japanese (Intermediate)

SKILLS

Programming Languages: Java, Kotlin, Python, C/C++, C#, JavaScript/Node.js

Web and App Development: React, Flask, FastAPI, Qt/PyQt, Swing, JavaFX, Unity, HTML, CSS, SASS, Bootstrap

Systems and Scripting: Bash (Linux), Zsh (macOS), Batch/PowerShell (Windows), Perl, AWK

Databases: Oracle, SQLite, Apache Derby

Cloud and DevOps: VMware, Docker, AWS, Git/GitHub/GitLab, CI/CD, Apache Tomcat

Software Engineering: OOP, MVC/MVVM, Design Patterns, Agile/Scrum

Developer Tools: Gradle, Maven, CMake, Make, Poetry, Doxygen, Javadoc, JSDoc

REFERENCES

Available upon request.